Section 2 / Module 1 / Math / Baseline

1

The function f is defined by the equation $f(x) = 100x + 7$. What is the value of $f(x)$ when $x = 5$?

A) 112

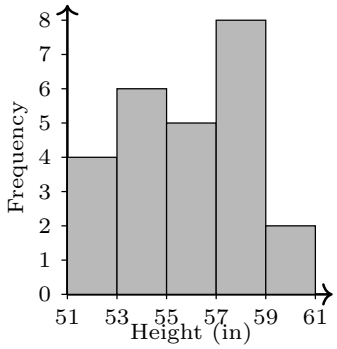
B) 135

C) 500

D) 507

2

For a project, Kayla recorded the heights of 25 people in her neighborhood. The histogram summarizes the heights, in inches (in), of these 25 people in Kayla's neighborhood. The first bar represents people with a height of at least 51 in but less than 53 in. The second bar represents people with a height of at least 53 in but less than 55 in, and so on.



How many of these 25 people in Kayla's neighborhood had a height of at least 51 in but less than 55 in?

Javob: _____

3

Which expression is equivalent to $32x^2 + 32$?

A) $32(x^2 + 1)$

B) $32(x^2 + 32)$

C) $32(32x^2 + 1)$

D) $32\left(x^2 + \frac{1}{32}\right)$

4

If $2(x - 10) = (2)(80)$, what is the value of $x - 10$?

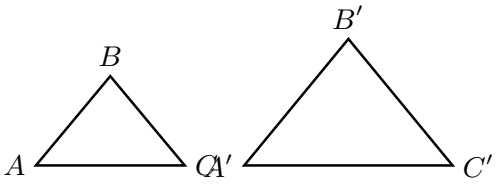
A) 8

B) 16

C) 80

D) 160

5



Note: Figures not drawn to scale.

Triangles ABC and $A'B'C'$ are shown. Triangle ABC is dilated by a scale factor of 3 to form triangle $A'B'C'$. If the length of $\overline{AB}$ is 21, what is the length of $\overline{A'B'}$?

A) 3

B) 7

C) 24

D) 63

6

What is an x-intercept of the graph of $y = x^2 - 81$ in the xy-plane?

A) $(1, 0)$

B) $(9, 0)$

C) $(18, 0)$

D) $(81, 0)$

7

A store sells two different-sized bottles of a certain shampoo. The store's sales of this shampoo totaled 910.53 dollars last month. The equation $4.59x+6.09y = 910.53$ represents this situation, where x is the number of smaller bottles sold and y is the number of larger bottles sold. According to the equation, what is the price, in dollars, of each smaller bottle?

Javob: _____

8

Triangle ABC is similar to triangle XYZ, where A, B, and C correspond to X, Y, and Z, respectively. In triangle ABC, the length of $\overline{AB}$ is 180 and the length of $\overline{BC}$ is 900. In triangle XYZ, the length of $\overline{YZ}$ is 80. What is the length of $\overline{XY}$?

A) 288

B) 196

C) 80

D) 16

9

In the xy-plane, line k passes through the points $(0, -9)$ and $(1, -5)$. Which equation defines line k?

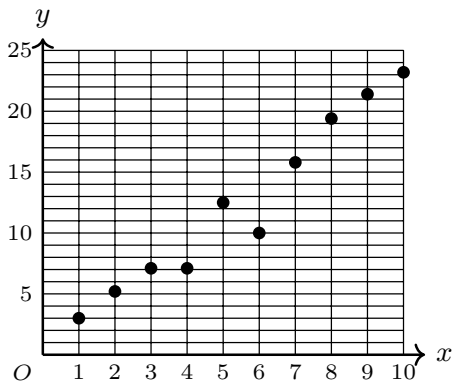
A) $y = \frac{1}{4}x - 9$

B) $y = -x + \frac{1}{4}$

C) $y = 4x - 9$

D) $y = -x + 4$

10



Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

A) $y = 0.2 + 2.28x$

B) $y = 0.2 - 2.28x$

C) $y = 2.28 + 0.2x$

D) $y = 2.28 - 0.2x$

11

$g(x) = (24 - 2x)(10 + 2x)$

The function g is defined by the given equation. For what value of x does $g(x)$ reach its maximum?

A) 3.5

B) 7

C) 8.5

D) 17

12

An exponential function f is defined by $f(x) = k(2)^x$, where k is a positive constant. If $f(-1) = 8$ and $f(2) = 64$, what is the value of k?

A) 8

B) 9

C) 15

D) 16

13

The base of a right rectangular pyramid has a length of 27 centimeters and a width of 7 centimeters. The height of this pyramid is 5 centimeters. What is the volume, in cubic centimeters, of this pyramid?

Javob: _____

14

$66 = \dfrac{2K}{v^2}$

For an object with a mass of 66 kilograms, the given equation relates the kinetic energy K, in joules, of the object to the object's speed v, in meters per second, where K and v are positive. Which equation correctly expresses this object's speed, in meters per second, in terms of the object's kinetic energy, in joules?

A) $v = \dfrac{2K}{66^2}$

B) $v = \dfrac{66^2}{2K}$

C) $v = \sqrt{\dfrac{66}{2K}}$

D) $v = \sqrt{\dfrac{2K}{66}}$

15

If $\frac{j}{k} = 95$ and $\frac{36j}{12xk} = 95$, what is the value of x?

A) $\frac{12}{95}$

B) $\frac{1}{3}$

C) 3

D) 36

16

$$\begin{cases} y \geq x + 15 \\ y \leq -5x - 15 \end{cases}$$

Which point (x, y) is a solution to the given system of inequalities in the xy-plane?

A) $(-6, 13)$

B) $(-6, 8)$

C) $(-4, 13)$

D) $(-4, 8)$

17

A plumbing company charges its customers \$111 for the first hour of a plumbing service call and \$50 for each additional hour, plus the cost of plumbing parts. Which equation represents the total cost y, in dollars, for an x-hour plumbing service call where the cost of parts was $\$z$ and $x \geq 1$?

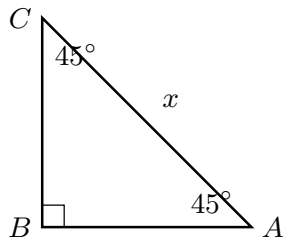
A) $y = 50x + z$

B) $y = 50x + z + 61$

C) $y = 111x + z$

D) $y = 111x + 50z + 61$

18



In the triangle shown, the value of x is 8. What is the value of $\sin A$?

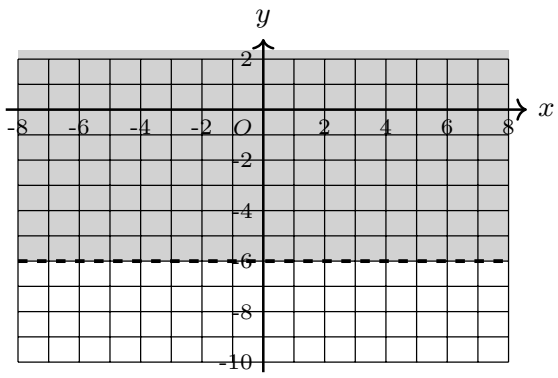
A) $\frac{1}{\sqrt{3}}$

B) $\frac{1}{\sqrt{2}}$

C) 1

D) $\sqrt{2}$

19



The shaded region shown represents the solutions to the inequality $-19y < c$, where c is a constant. What is the value of c?

A) 114

B) 6

C) -6

D) -114

20

$$\begin{cases} y = x - 1 \\ y = 27(x-1)^2 - 28 \end{cases}$$

A solution to the given system of equations is (x, y). What is one possible value of $x - 1$?

A) -2

B) $-\dfrac{28}{27}$

C) -1

D) $-\dfrac{1}{27}$

21

$$\begin{cases} 10x + 14y = 6y + 5 \\ ty = \dfrac{1}{3} + 5x \end{cases}$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t?

Javob: _____

22

$4|x + 3| - 2 = 22$

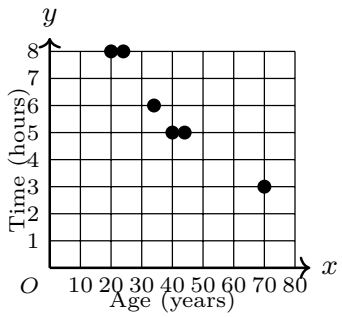
What is the sum of the solutions to the given equation?

Javob: _____

Section 2 / Module 2 / Math / Easy

23

The scatterplot shows the relationship between the age, in years, of 6 people and the average amount of time, in hours, spent by each of these 6 people on the internet per day.



What is the average amount of time, in hours, spent by the 70-year-old person on the internet per day?

A) 3

B) 5

C) 7

D) 8

24

$x + 31 = 62$

What is the solution to the given equation?

A) 2

B) 31

C) 62

D) 93

25

In a litter of 5 kittens, 2 kittens are striped. If a kitten in this litter is selected at random, what is the probability of selecting a kitten that is striped?

A) $\frac{1}{5}$

B) $\frac{2}{5}$

C) $\frac{3}{5}$

D) $\frac{5}{5}$

26

What is the y-intercept of the graph of $y = \left(\frac{2}{17}\right)^{x+1}$ in the xy-plane?

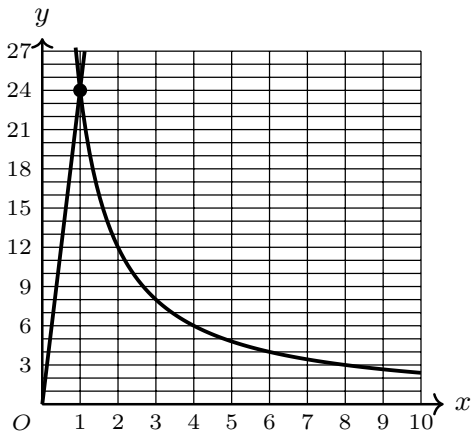
A) $(2, 17)$

B) $(0, 17)$

C) $(0, 2)$

D) $\left(0, \frac{2}{17}\right)$

27



The system of two equations, $y = \dfrac{24}{x}$ and $y = 24x$, is graphed in the xy-plane as shown. Which ordered pair is a solution to this system?

A) $(0, 0)$

B) $(0, 24)$

C) $(1, 24)$

D) $(24, 24)$

28

A certain pigeon species can fly at an average speed of 16 meters per second when in continuous flight. At this rate, how many meters would this pigeon species fly in 3 seconds?

A) 48

B) 19

C) 16

D) 13

29

$x + y = 92$

The given equation relates the total number of marble queen pothos stems, x, and the total number of snow queen pothos stems, y, a plant nursery propagates to create pothos plants. If 32 snow queen pothos stems are propagated, how many marble queen pothos stems are propagated?

A) 92

B) 60

C) 32

D) 28

30

A shipment consists of 5-pound boxes and 10-pound boxes with a total weight of 330 pounds. There are 15 10-pound boxes in the shipment. How many 5-pound boxes are in the shipment?

A) 5

B) 10

C) 15

D) 36

31

Which expression is equivalent to $14x^2 - 5x^2$?

A) $-70x^2$

B) $9x^2$

C) $19x^2$

D) $38x^2$

32

In triangle ABC, the measure of angle A is $36°$ and the measure of angle B is $90°$. What is the measure, in degrees, of angle C?

A) 36

B) 54

C) 90

D) 180

33

A shopper goes to a pet store to buy dog food and treats. The shopper has \$15. Each can of food costs \$1.73, and each bag of treats costs \$2.22. Which inequality represents the relationship between the number of cans of food, c, and the number of bags of treats, t, the shopper can buy?

A) $1.73c + 2.22t \leq 15$

B) $1.73c + 2.22t \geq 15$

C) $2.22c + 1.73t \leq 15$

D) $2.22c + 1.73t \geq 15$

34

Out of 500 attempts, 55% were successful. How many of these attempts were successful?

Javob: _____

35

The function f is defined by $f(x) = x^2 + x + 53$. What is the value of $f(2)$?

Javob: _____

36

$y = 3{,}100(a)^x$

The given equation, where a is a positive constant, gives the predicted number of bacteria, y, in a growth medium x hours after the number of bacteria was initially measured. According to the equation, what was the predicted number of bacteria initially measured in the growth medium?

Javob: _____

37

Triangle R has an area of 40 square centimeters (cm^2). Square S has side lengths of 4 cm. What is the total area of triangle R and square S, in cm^2?

A) 22

B) 24

C) 44

D) 56

38

$$\begin{cases} 3x - 3y = 9 \\ -8y = 32 \end{cases}$$

The solution to the given system of equations is (x, y). What is the value of x?

Javob: _____

39

What is an x-intercept of the graph of the equation $y = (x + 7)(x - 9)(x + 12)$ in the xy-plane?

A) $(3, 0)$

B) $(0, 7)$

C) $(9, 0)$

D) $(0, 12)$

40

x	$f(x)$
0	7
1	13
2	19

For a linear function f, the table shows three values of x and their corresponding values of $f(x)$. Which of the following could define f?

A) $f(x) = 7x + 6$

B) $f(x) = 6x + 7$

C) $f(x) = x + 7$

D) $f(x) = 6x$

41

A car dealership has only sedans, SUVs, and minivans for sale. On Monday, 20% of the vehicles for sale were sedans and 50% were SUVs. If there were 22 sedans for sale at the dealership on Monday, how many minivans were for sale?

Javob: _____

42

$x^2 = -9$

How many distinct real solutions does the given equation have?

A) Exactly one

B) Exactly two

C) Infinitely many

D) Zero

43

A circle in the xy-plane has the equation $(x-17)^2+(y-k)^2=36$. Which of the following gives the center of the circle and its radius?

A) The center is at $(17, k)$ and the radius is 6.

B) The center is at $(k, 17)$ and the radius is 6.

C) The center is at $(17, k)$ and the radius is 36.

D) The center is at $(k, 17)$ and the radius is 36.

44

In the xy-plane, line k passes through the points $(3, 0)$ and $(0, 5)$. Which of the following is an equation of line k?

A) $5x + 3y = 15$

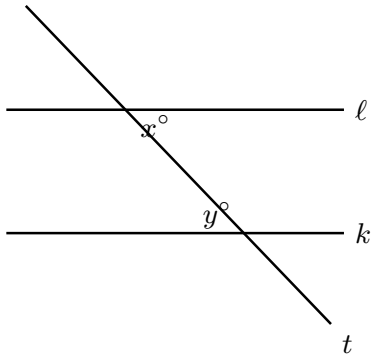
B) $3x + 5y = 15$

C) $5x - 3y = 15$

D) $3x - 5y = 15$

Section 2 / Module 2 / Math / Hard

45



Note: Figure not drawn to scale.

In the figure shown, lines ℓ and k are parallel and line t intersects both lines. If $x > 120$, which of the following must be true?

A) $y < 60$

B) $y > 60$

C) $x + y < 180$

D) $x + y > 180$

46

Area (square feet)	Water (gallons)
1,040	2,496
1,560	3,744
2,080	4,992

The roofs on the buildings of a public garden are designed to allow water to drain into the garden. The table shows the relationship between the area x, in square feet, of a roof and the amount of water $f(x)$, in gallons, drained from the roof into the garden over a certain period of time. Which equation could define f?

A) $f(x) = 0.6x$

B) $f(x) = 2.4x$

C) $f(x) = 1{,}248.0x$

D) $f(x) = 2{,}496.0x$

47

$13x + 325 = a(x + 25)$

In the given equation, a is a constant. If the equation has infinitely many solutions, what is the value of a?

Javob: _____

48

$$\begin{cases} y = -\dfrac{1}{11}x \\ y = \dfrac{1}{4}x \end{cases}$$

The solution to the given system of equations is (x, y). What is the value of x?

A) -11

B) -7

C) 0

D) 4

49

An organization sells tickets to a fundraiser for \$50 per person. The organization's expenses are \$600 for a rental space and \$18 per person for food. The organization's goal is to earn at least \$1,500 after paying all expenses. What is the minimum number of tickets that must be sold to meet this goal?

A) 30

B) 42

C) 47

D) 66

50

$f(x) = \dfrac{a-15}{x} + 5$

In the given function f, a is a constant. The graph of function f in the xy-plane, where $y = f(x)$, is translated 3 units down and 2 units to the right to produce the graph of $y = g(x)$. Which equation defines function g?

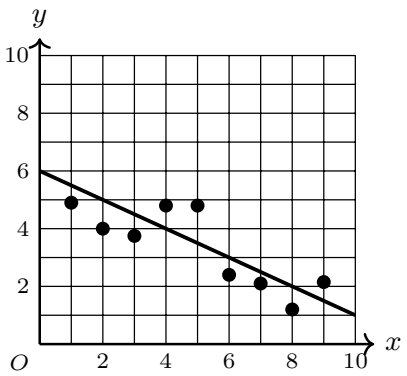
A) $g(x) = \dfrac{a-15}{x+2} + 2$

B) $g(x) = \dfrac{a-15}{x-2} + 2$

C) $g(x) = \dfrac{a-18}{x+2} + 5$

D) $g(x) = \dfrac{a-18}{x-2} + 5$

51



The scatterplot shows the relationship between two variables, x and y. A line of best fit for the data is also shown. For how many of the 9 data points does the line of best fit predict a y-value that is less than the actual y-value?

A) 9

B) 6

C) 3

D) 0

52

Line p is defined by $2y + 18x = 11$. Line r is perpendicular to line p in the xy-plane. What is the slope of line r?

A) 9

B) $\frac{1}{9}$

C) $-\frac{1}{9}$

D) -9

53

What is the y-intercept of the graph of $y = \left(\frac{2}{17}\right)^{x+1}$ in the xy-plane?

A) $(2, 17)$

B) $(0, 17)$

C) $(0, 2)$

D) $\left(0, \frac{2}{17}\right)$

54

A beaker containing a liquid is placed on a table. The function

$$g(t) = 294 + (364 - 294)(2.72)^{-0.104t}$$

gives the approximate temperature, in kelvins, of the liquid t minutes after the beaker was placed on the table. According to this function, what was the approximate temperature, in kelvins, of the liquid when the beaker was placed on the table?

Javob: _____

55

Hannah and Wyatt are saving money to purchase a car. Hannah saves $\frac{1}{5}$ of her salary each month, and Wyatt saves $\frac{2}{7}$ of his salary each month. Together they save a total of \$3,160 from their monthly salaries each month. If h and w represent Hannah's and Wyatt's monthly salaries, in dollars, respectively, which equation shows the relationship between h and w?

A) $h + w = 3{,}160$

B) $h + 2w = 3{,}160$

C) $10h + 7w = 110{,}600$

D) $7h + 10w = 110{,}600$

56

56 SAVOL

$f(x) = \sqrt{7x + 6}$

The function f is defined by the given equation. If $f(a) = -7a$, where a is a constant, what is the value of a?

A) $-\frac{3}{7}$

B) $-\frac{2}{7}$

C) $\frac{2}{7}$

D) $\frac{3}{7}$

57

A car dealership has only sedans, SUVs, and minivans for sale. On Monday, 20% of the vehicles for sale were sedans and 50% were SUVs. If there were 22 sedans for sale at the dealership on Monday, how many minivans were for sale?

Javob: _____

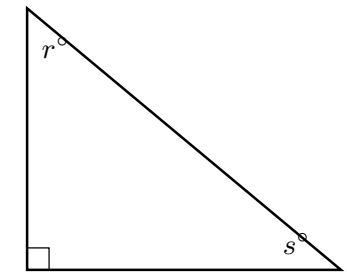
58

$4x^2 + 14 = 32x + r$

In the given equation, r is a constant. The equation has exactly one real solution. What is the value of r?

Javob: _____

59



Note: Figure not drawn to scale.

In the figure, $3r > 11s$. Which of the following values must be the greatest?

A) $\sin r^\circ$

B) $\cos r^\circ$

C) $\dfrac{\sin r^\circ}{\sin s^\circ}$

D) $\dfrac{\sin r^\circ}{\cos s^\circ}$

60

A researcher conducted a study to examine the melatonin levels of gamers and the amount of time the gamers spent playing video games. A total of 127 gamers were selected at random from three countries. The melatonin levels of the gamers and the amount of time they spent playing video games were recorded over a 4-week period. The table gives the number of selected gamers from each country.

Country	Sweden	South Korea	United States
Number of gamers	25	49	53

If the results of this study suggest a relationship between melatonin levels and the amount of time spent playing video games, which of the following aspects of the study will prevent the results from being generalized to all gamers?

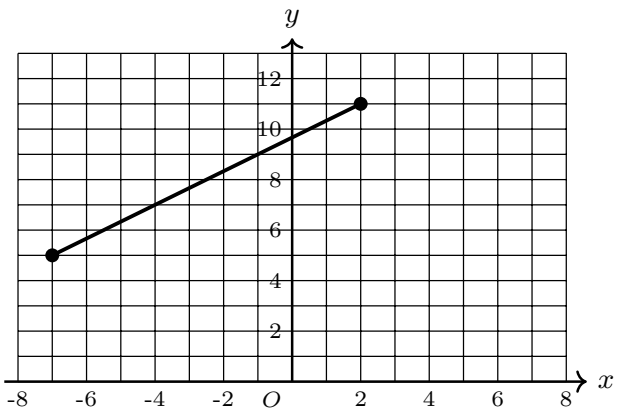
A) The melatonin levels of the selected gamers were not measured over a long enough period of time.

B) The number of selected gamers from each country is different.

C) The number of selected gamers is too small.

D) The gamers were not selected at random from all gamers.

61



The line segment shown in the xy-plane represents one of the legs of a right triangle. The area of this triangle is $42\sqrt{13}$ square units. What is the length, in units, of the other leg of this triangle?

Javob: _____

62

$-12(5x-3)^2+6(5x-4)^2$

The given expression can be rewritten as $\frac{a}{6}x^2+\frac{b}{6}x+\frac{c}{6}$, where a, b, and c are constants. What is the value of $a+b+c$?

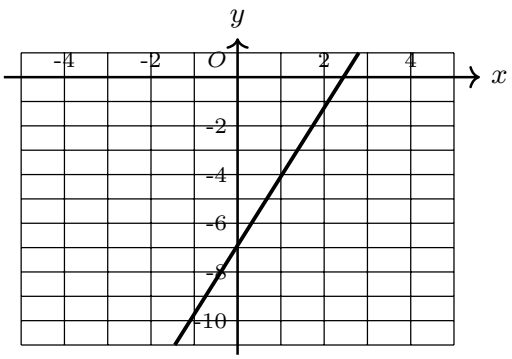
A) -252

B) -126

C) -42

D) -7

63



The graph of the linear function $y = g(x) - 17$ is shown. If a and b are positive constants, which equation could define g?

A) $g(x) = -ax + b$

B) $g(x) = ax - b$

C) $g(x) = ax + b$

D) $g(x) = -ax - b$

64

A circle has center G, and points M and N lie on the circle. Line segments $\overline{MH}$ and $\overline{NH}$ are tangent to the circle at points M and N, respectively. If the radius of the circle is 238 millimeters and the perimeter of quadrilateral $GMHN$ is 4,508 millimeters, what is the distance, in millimeters, between points G and H?

A) 238

B) 2,002

C) 2,016

D) 2,030

65

$h(x) = a(x + 43)(x + r)(x + s)$

The polynomial function h is defined by the given equation, where a, r, and s are positive constants. In the xy-plane, the graph of $y = h(x)$ intersects the x-axis at exactly three points. If $h(0) = p$, which of the following must be true?

A) $p < 43rs$

B) $p = 43rs$

C) $p > 43rs$

D) There is not enough information to compare the values of p and $43rs$.

66

The mass of object A is 462% of the mass of object B, and the mass of object A is 0.084% of the mass of object C. If the mass of object C is p% of the mass of object B, what is the value of $\frac{p}{1{,}000}$?

Javob: _____

67

The function h is defined by $h(x) = 100x - 50$. What is the value of $h(x)$ when $x = 6$?

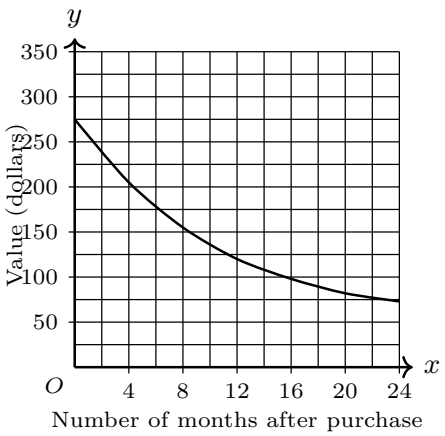
A) 56

B) 156

C) 550

D) 650

68



The graph shown gives the estimated value, in dollars, of a mountain bike as a function of the number of months since it was purchased. What is the best interpretation of the y-intercept of the graph in this context?

A) The estimated value of the bike was \$275 when it was purchased.

B) The estimated value of the bike 24 months after it was purchased was \$275.

C) The estimated value of the bike had decreased by \$275 in the 24 months after it was purchased.

D) The estimated value of the bike decreased by approximately 2.75% each year after it was purchased.

69

Out of 200 attempts, 55% were successful. How many of these attempts were successful?

Javob: _____

70

If $2(x-3) = (2)(90)$, what is the value of $x-3$?

A) 30

B) 60

C) 90

D) 180

71

Which expression is equivalent to $29x^2+29$?

A) $29(x^2+1)$

B) $29(x^2+29)$

C) $29(29x^2+1)$

D) $29\left(x^2+\frac{1}{29}\right)$

72

A commercial water pump drains water from a reservoir at a constant rate of 180 gallons per minute. At this rate, how many hours will it take to drain 32,400 gallons of water?

A) 2.5

B) 3.0

C) 3.5

D) 4.0

73

A store sells two different-sized bottles of a certain shampoo. The store's sales of this shampoo totaled 902.07 dollars last month. The equation $4.57x+6.08y = 902.07$ represents this situation, where x is the number of smaller bottles sold and y is the number of larger bottles sold. According to the equation, what is the price, in dollars, of each smaller bottle?

Javob: _____

74

Triangle ABC is similar to triangle XYZ, where A, B, and C correspond to X, Y, and Z, respectively. In triangle ABC, the length of $\overline{AB}$ is 180 and the length of $\overline{BC}$ is 900. In triangle XYZ, the length of $\overline{YZ}$ is 80. What is the length of $\overline{XY}$?

A) 288

B) 196

C) 80

D) 16

75

$|x| - 43 = 0$

Which value is a solution to the given equation?

A) $\frac{1}{43}$

B) 0

C) $-\frac{1}{43}$

D) -43

76

In the xy-plane, line k passes through the points $(0, -8)$ and $(1, -5)$. Which equation defines line k?

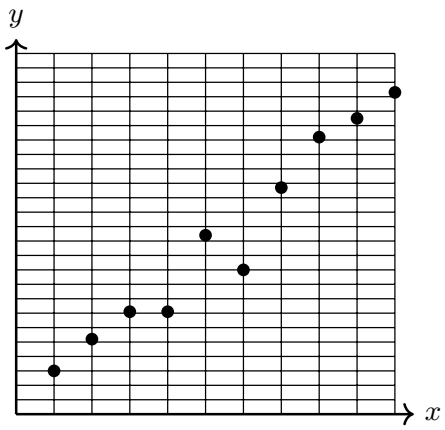
A) $y = \frac{1}{3}x - 8$

B) $y = -x + \frac{1}{3}$

C) $y = 3x - 8$

D) $y = -x + 3$

77



Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

A) $y = 0.2 + 2.25x$

B) $y = 0.2 - 2.25x$

C) $y = 2.25 + 0.2x$

D) $y = 2.25 - 0.2x$

78

What is the x-intercept of the graph of $y = \dfrac{21 - x}{ax + b}$ in the xy-plane, where a and b are positive constants?

A) $(21, 0)$

B) $(0, -21)$

C) $\left(0, \dfrac{21}{b}\right)$

D) $\left(-\dfrac{b}{a}, 0\right)$

79

A model estimates that a glacier moves between 17 and 24 centimeters per day. Which inequality represents possible values of x, where x is the total distance, in centimeters, the glacier moves in 2 days?

A) $0 \leq x \leq 2$

B) $(17)(0.5) \leq x \leq (24)(0.5)$

C) $17 \leq x \leq 24$

D) $(17)(2) \leq x \leq (24)(2)$

80

In the ancient Roman measurement system, a leuga was a unit of length that was equal to 7,500 pedes. In this measurement system, how many pedes were equivalent to 4.6 leugas?

Javob: _____

81

A rectangle has a width of n yards and a length of $(n + 7)$ yards. If the rectangle has an area of 198 square yards, what is the value of n?

Javob: _____

82

$55 = \dfrac{2K}{v^2}$

For an object with a mass of 55 kilograms, the given equation relates the kinetic energy K, in joules, of the object to the object's speed v, in meters per second, where K and v are positive. Which equation correctly expresses this object's speed, in meters per second, in terms of the object's kinetic energy, in joules?

A) $v = \dfrac{2K}{55^2}$

B) $v = \dfrac{55^2}{2K}$

C) $v = \sqrt{\dfrac{55}{2K}}$

D) $v = \sqrt{\dfrac{2K}{55}}$

83

If $\dfrac{j}{k} = 83$ and $\dfrac{44j}{22xk} = 83$,
what is the value of x?

A) $\dfrac{22}{83}$

B) $\dfrac{1}{2}$

C) 2

D) 44

84

A certain company measures its solar cell production $C(n)$, in peak kilowatts, generated by all the solar cells manufactured by the company. The company's solar cell production grew by approximately 22% each year for the 5 years following 2013. If solar cell production was 7,138 peak kilowatts in 2013, which of the following functions best models the solar cell production, in peak kilowatts, n years after 2013, where $0 \leq n \leq 5$?

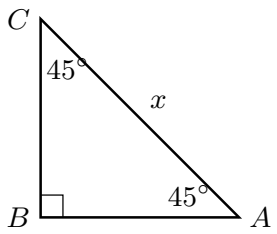
A) $C(n) = 7{,}138(0.22)^n$

B) $C(n) = 7{,}138(1.22)^n$

C) $C(n) = 0.22(7{,}138)^n$

D) $C(n) = 1.22(7{,}138)^n$

85



In the triangle shown, the value of x is 19. What is the value of $\sin A$?

A) $\frac{1}{\sqrt{3}}$

B) $\frac{1}{\sqrt{2}}$

C) 1

D) $\sqrt{2}$

86

A right square prism has a height of 15 units. The volume of the prism is 1,815 cubic units. What is the length, in units, of an edge of the base?

Javob: _____

87

$$\begin{cases} y = x - 1 \\ y = 19(x-1)^2 - 20 \end{cases}$$

A solution to the given system of equations is (x, y). What is one possible value of $x - 1$?

A) -2

B) $-\dfrac{20}{19}$

C) -1

D) $-\dfrac{1}{19}$

88

$$\begin{cases} 18x + 6y = 8y + 9 \\ ty = \dfrac{1}{5} + 9x \end{cases}$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t?

Javob: _____

89

A circle in the xy-plane has center $(24, 5)$ and y-intercepts $(0, -27)$ and $(0, 37)$. What is the radius of this circle?

A) 40

B) 37

C) 32

D) 27

90

For $x > 0$, which expression is equivalent to

$$\sqrt{x^3} \cdot \sqrt[4]{x}\,?$$

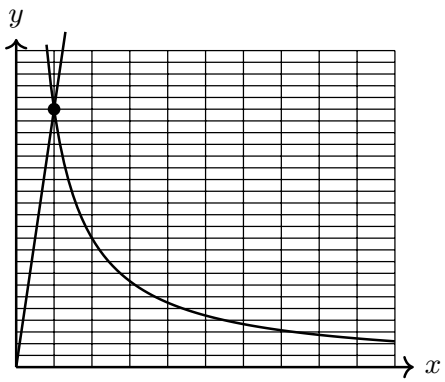
A) $x^{3/8}$

B) $x^{7/4}$

C) $x^{5/4}$

D) $x^{3/4}$

91



The system of two equations, $y = \dfrac{22}{x}$ and $y = 22x$, is graphed in the xy-plane as shown. Which ordered pair is a solution to this system?

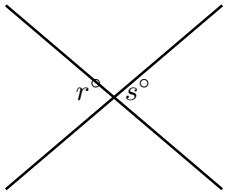
A) $(0, 0)$

B) $(0, 22)$

C) $(1, 22)$

D) $(22, 22)$

92



Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. If $r = 71$, what is the value of s?

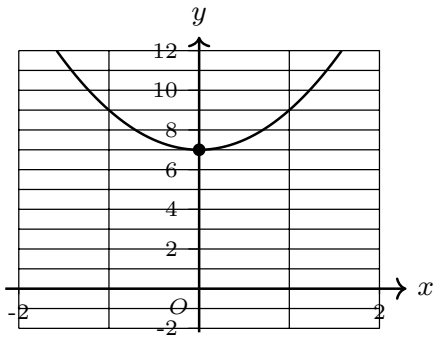
A) 19

B) 71

C) 161

D) 251

93



The y-intercept of the graph shown is (x, y). What is the value of y?

Javob: _____

94

$x + y = 88$

The given equation relates the total number of marble queen pothos stems, x, and the total number of snow queen pothos stems, y, a plant nursery propagates to create pothos plants. If 28 snow queen pothos stems are propagated, how many marble queen pothos stems are propagated?

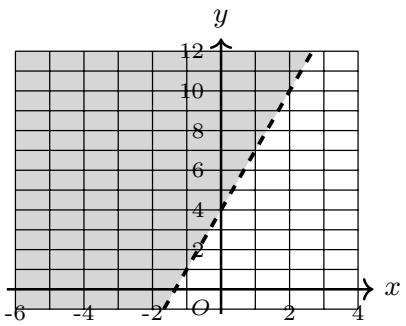
A) 88

B) 60

C) 32

D) 28

95



The shaded region shown represents the solutions to which inequality?

A) $y < 3x + 4$

B) $y < 3x - 4$

C) $y > 3x + 4$

D) $y > 3x - 4$

96

If $3(2x - 5) = 4x + 7$, what is the value of x?

A) 8

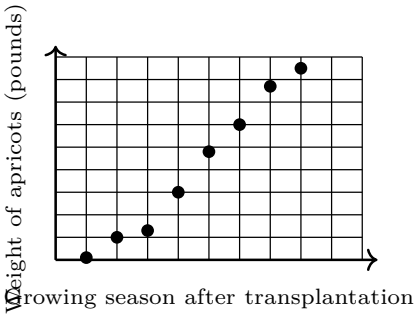
B) 11

C) 14

D) 22

97

A botanist recorded the weight, in pounds, of all apricots that grew on a dwarf apricot tree during each growing season after the tree's transplantation. The scatterplot shows this weight, in pounds, for each growing season after the tree's transplantation.



What was the weight, to the nearest pound, of all apricots that grew on the tree during the 4th growing season after the tree's transplantation?

Javob: _____

98

A shipment consists of 5-pound boxes and 10-pound boxes with a total weight of 300 pounds. There are 13 10-pound boxes in the shipment. How many 5-pound boxes are in the shipment?

A) 5

B) 10

C) 13

D) 34

99

Which expression is equivalent to $18x^2 - 3x^2$?

A) $-54x^2$

B) $15x^2$

C) $21x^2$

D) $42x^2$

100

A data set consists of 8 integers: $4, 7, 7, 9, 12, 14, 15, 20$. What is the median of this data set?

A) 7

B) 9

C) 10.5

D) 11

101

$4(x-5)+8=2(x-3)+26$

What value of x is the solution to the given equation?

A) 56

B) 32

C) 28

D) 16

102

A market researcher collected data on the price of an iced coffee at 42 locations selected at random in a state. The mean price of an iced coffee in the state was estimated to be \$1.37, with an associated margin of error of \$0.05. Which of the following is a plausible statement about the mean price of an iced coffee for all locations that sell this product in the state?

A) It is between \$1.32 and \$1.42.

B) It is either less than \$1.32 or greater than \$1.42.

C) It is less than \$1.32.

D) It is greater than \$1.42.

103

A teacher is creating an assignment worth 10 points. The assignment will consist of questions worth 1 point and questions worth 2 points. Which equation represents this situation, where x represents the number of 1-point questions and y represents the number of 2-point questions?

A) $3xy = 10$

B) $3(x+y) = 10$

C) $2x + y = 10$

D) $x + 2y = 10$

104

Data set X: $8, 12, 12, 16$ Data set Y: $8, 12, 12, 16, 33$

The lists give the values in data sets X and Y. Which statement correctly compares the mean of data set X and the mean of data set Y?

A) The mean of data set X is greater than the mean of data set Y.

B) The mean of data set X is less than the mean of data set Y.

C) The means of data set X and data set Y are equal.

D) There is not enough information to compare the means.

105

The function f is defined by $f(x) = x^3 + 26$. What is the value of $f(2)$?

Javob: _____

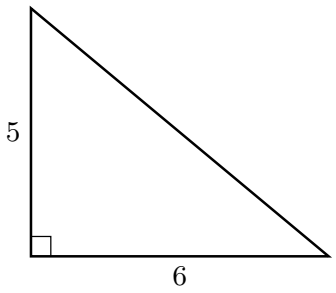
106

$y = 2{,}200(a)^x$

The given equation, where a is a positive constant, gives the predicted number of bacteria, y, in a growth medium x hours after the number of bacteria was initially measured. According to the equation, what was the predicted number of bacteria initially measured in the growth medium?

Javob: _____

107



Note: Figure not drawn to scale.

The figure shows the lengths, in units, of two sides of a triangle. What is the area, in square units, of the triangle?

A) 15

B) 11

C) 6

D) 5

108

What is the positive solution to the equation $x^2 - 7x - 18 = 0$?

A) 2

B) 6

C) 9

D) 18

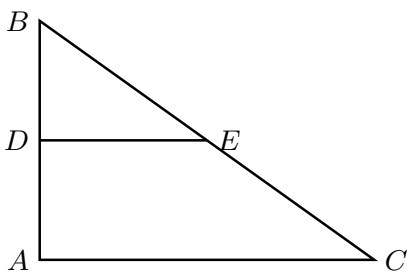
109

$$\begin{cases} 2x - 2y = 8 \\ -3y = 18 \end{cases}$$

The solution to the given system of equations is (x, y). What is the value of x?

Javob: _____

110



Note: Figure not drawn to scale.

In triangle ABC, $\overline{DE}$ is parallel to $\overline{AC}$, $BD = AD$, $BE = 9$, and $AC = 8$. What is the length of $\overline{BC}$?

A) 72

B) 18

C) 17

D) 16

111

x	y
-4	11
0	12
8	14

The table shows three values of x and their corresponding values of y. There is a linear relationship between x and y. Which equation represents this relationship?

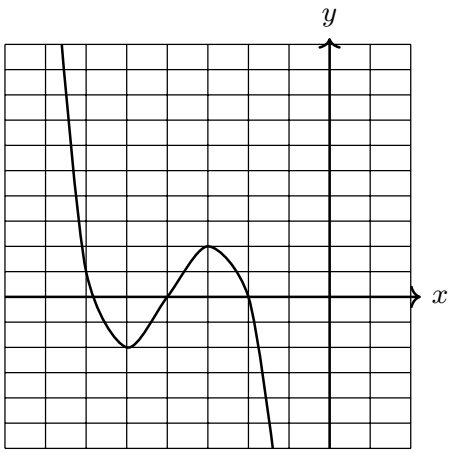
A) $y = \frac{1}{4}x + 12$

B) $y = \frac{1}{12}x + 4$

C) $y = 4x + 12$

D) $y = 12x + 4$

112



The graph of the function f is shown, where $y = f(x)$. What is the value of $f(-3)$?

A) -5

B) -3

C) 0

D) 2

113

Line q is defined by $y = -\frac{1}{7}x - \frac{1}{2}$. Line r is perpendicular to line q in the xy-plane. What is the slope of line r?

A) 9

B) 7

C) 5

D) 2

114

$y > 5x + 7$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A)

x	y
0	8
1	13
2	18

B)

x	y
0	7
1	12
2	17

C)

x	y
0	6
1	11
2	16

D)

x	y
0	5
1	6
2	7

115

$x^2 - \frac{25}{9} = 0$

How many distinct real solutions does the given equation have?

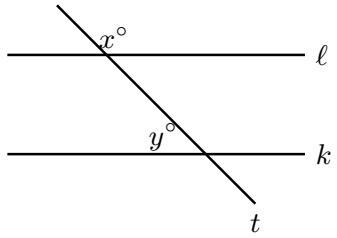
A) Zero

B) Exactly one

C) Exactly two

D) Infinitely many

116



Note: Figure not drawn to scale.

In the figure shown, lines ℓ and k are parallel and line t intersects both lines. If $x > 137$, which of the following must be true?

A) $y < 43$

B) $y > 43$

C) $x + y < 180$

D) $x + y > 180$

117

A circle in the xy-plane has equation $(x - 4)^2 + (y + 1)^2 = 49$. What is the radius of the circle?

A) 7

B) 14

C) 24.5

D) 49

118

Area (square feet)	Water (gallons)
2,400	4,320
3,600	6,480
4,800	8,640

The roofs on the buildings of a public garden are designed to allow water to drain into the garden. The table shows the relationship between the area x, in square feet, of a roof and the amount of water $f(x)$, in gallons, drained from the roof into the garden over a certain period of time. Which equation could define f?

A) $f(x) = 0.6x$

B) $f(x) = 1.8x$

C) $f(x) = 2{,}160.0x$

D) $f(x) = 4{,}320.0x$

119

$15x + 435 = a(x + 29)$

In the given equation, a is a constant. If the equation has infinitely many solutions, what is the value of a?

Javob: _____

120

What is the slope of the graph of $y = \frac{1}{3}(26x + 11) + 9x$ in the xy-plane?

A) $\frac{26}{3}$

B) $\frac{53}{3}$

C) 26

D) 35

121

A plumbing company charges its customers \$116 for the first hour of a plumbing service call and \$70 for each additional hour, plus the cost of plumbing parts. Which equation represents the total cost y, in dollars, for an x-hour plumbing service call where the cost of parts was \$$z$ and $x \geq 1$?

A) $y = 70x + z$

B) $y = 70x + z + 46$

C) $y = 116x + z$

D) $y = 116x + 70z + 46$

122

x	25	27	29
$f(x)$	-4	-4	4

For the quadratic function f, the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f?

A) $f(x) = (x - 25)(x - 27) + 2$

B) $f(x) = (x - 25)(x - 27) - 4$

C) $f(x) = (x - 4)(x - 29) + 16$

D) $f(x) = (x - 4)(x - 29) - 16$

123

The function h is defined by $h(x) = |x - 8| - |6 - x| + a$, where a is a constant, and $h(-4) = 1$. What is the value of a?

Javob: _____

124

Line h is defined by $\frac{1}{3}x + \frac{1}{2}y - 30 = 0$. Line j is perpendicular to line h in the xy-plane. What is the slope of line j?

A) $-\frac{2}{3}$

B) $-\frac{3}{2}$

C) $\frac{2}{3}$

D) $\frac{3}{2}$

125

A laboratory culture of bacteria grows according to the model $P(t) = 250(1.08)^t$, where t represents elapsed time in hours. By what percentage does the population increase each hour?

A) 0.08%

B) 8%

C) 25%

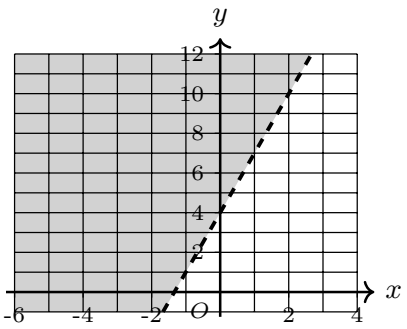
D) 108%

126

A beaker containing a liquid is placed on a table. The function $g(t) = 296 + (362 - 296)(2.72)^{-0.104t}$ gives the approximate temperature, in kelvins, of the liquid t minutes after the beaker was placed on the table. According to this function, what was the approximate temperature, in kelvins, of the liquid when the beaker was placed on the table?

Javob: _____

127



The shaded region shown represents the solutions to which inequality?

A) $y < 3x + 4$

B) $y < 3x - 4$

C) $y > 3x + 4$

D) $y > 3x - 4$

128

$$\begin{cases} 3(2x) + 8(9y) = 534 \\ 3(2x) - 8(9y) = -618 \end{cases}$$

The solution to the given system of equations is (x, y). What is the value of $2x + 9y$?

A) -86

B) -14

C) 58

D) 72

129

The function f is defined by $f(x) = -2(x-3)^2 + 8$. What is the maximum value of $f(x)$?

A) -2

B) 3

C) 6

D) 8

130

The frequency table summarizes the age, in years, of each of the 50 coins in Vivek's original collection.

Age (years)	Frequency
60	10
65	5
70	4
75	12
80	4
85	5
90	10
Total	50

Vivek will add a 130-year-old coin to his original collection, creating a new collection of 51 coins. Which statement correctly compares the mean ages and median ages of Vivek's original and new collections?

A) The mean age for the new collection is greater than the mean age for the original collection, and the median age for the new collection is greater than the median age for the original collection.

B) The mean age for the new collection is greater than the mean age for the original collection, but the median age is the same for the original and new collections.

C) The mean age for the new collection is less than the mean age for the original collection, and the median age for the new collection is less than the median age for the original collection.

D) The median age for the new collection is greater than the median age for the original collection, but the mean age is the same for the original and new collections.

131

$$\begin{cases} x^2 + y^2 = 64 \\ y = mx + \dfrac{b}{2} \end{cases}$$

In the given system of equations, m and b are negative constants. In the xy-plane, the graphs of the equations in the given system intersect at the point $(-7, y)$, where $y < 0$. Which expression represents the value of b?

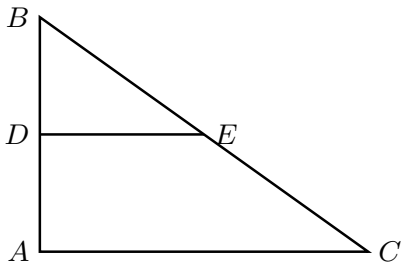
A) $14m - 2\sqrt{15}$

B) $-14m + 2\sqrt{15}$

C) $\dfrac{7m}{2} - \dfrac{\sqrt{15}}{2}$

D) $-\dfrac{7m}{2} + \dfrac{\sqrt{15}}{2}$

132



Note: Figure not drawn to scale.

In triangle ABC, $\overline{DE}$ is parallel to $\overline{AC}$, $BD = AD$, $BE = 9$, and $AC = 8$. What is the length of $\overline{BC}$?

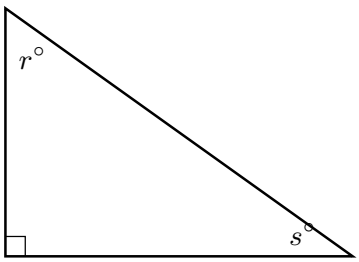
A) 72

B) 18

C) 17

D) 16

133



Note: Figure not drawn to scale.

In the figure, $3r > 19s$. Which of the following values must be the greatest?

A) $\sin r^\circ$

B) $\cos r^\circ$

C) $\dfrac{\sin r^\circ}{\sin s^\circ}$

D) $\dfrac{\sin r^\circ}{\cos s^\circ}$

134

Consider the system of equations below:

$$\begin{cases} 2x + 3y = 19 \\ x - y = 2 \end{cases}$$

What is the value of $x + y$?

A) 5

B) 7

C) 8

D) 11

135

A researcher conducted a study to examine the melatonin levels of gamers and the amount of time the gamers spent playing video games. A total of 125 gamers were selected at random from three countries. The melatonin levels of the gamers and the amount of time they spent playing video games were recorded over a 4-week period. The table gives the number of selected gamers from each country.

Country	Sweden	South Korea	United States
Number of gamers	25	49	51

If the results of this study suggest a relationship between melatonin levels and the amount of time spent playing video games, which of the following aspects of the study will prevent the results from being generalized to all gamers?

A) The gamers were not selected at random from all gamers.

B) The number of selected gamers is too small.

C) The number of selected gamers from each country is different.

D) The melatonin levels of the selected gamers were not measured over a long enough period of time.

136

In right triangle $\triangle ABC$, angle C measures $90°$. If $\cos(A) = \frac{5}{13}$, what is the value of $\tan(A)$?

A) $\frac{12}{5}$

B) $\frac{5}{12}$

C) $\frac{12}{13}$

D) $\frac{13}{5}$

137

At a hotel, guests used the pool for a total of t hours in March. At the same hotel, guests used the pool for $3.41t$ hours in April. What is the percent increase in the number of hours guests used the pool from March to April?

A) 2.41%

B) 3.41%

C) 241%

D) 341%

138

$-14(3x-4)^2+6(3x-2)^2$

The given expression can be rewritten as $\frac{a}{4}x^2+\frac{b}{4}x+\frac{c}{4}$, where a, b, c are constants. What is the value of $a+b+c$?

A) -32

B) -16

C) -8

D) -2

139

The sum of three consecutive even integers is less than 123. The least of the three integers is x. What is the greatest possible value of x?

A) 41

B) 40

C) 39

D) 38

140

Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is 52 square centimeters (cm^2), and the surface area of right rectangular prism Y is 832 cm^2. The volume of right rectangular prism X is 10.5 cubic centimeters (cm^3). What is the volume, in cm^3, of right rectangular prism Y?

Javob: _____

141

$18(w - a)(w + 5) = 6(w + 5)$

In the given equation, a is a constant. The sum of the solutions to the equation is $\frac{37}{3}$. What is the value of a?

Javob: _____

142

$h(x) = a(x + 39)(x + r)(x + s)$

The polynomial function h is defined by the given equation, where a, r, and s are positive constants. In the xy-plane, the graph of $y = h(x)$ intersects the x-axis at exactly three points. If $h(0) = p$, which of the following must be true?

A) $p < 39rs$

B) $p = 39rs$

C) $p > 39rs$

D) There is not enough information to compare the values of p and $39rs$.

143

A rectangle has a perimeter of 166 centimeters and a length of 48 centimeters. What is the area, in square centimeters, of the rectangle?

A) 5,664

B) 2,304

C) 83

D) 1,680

144

The function f is defined by $f(x) = 9x - 13$. For what value of x does $f(x) = 437$?

Javob: _____

145

A bag contains 20 cards, each labeled with a different whole number from 1 through 20. If one card is selected at random, what is the probability that the number on the card is a multiple of 3 or a multiple of 4?

A) $\frac{3}{10}$

B) $\frac{1}{2}$

C) $\frac{11}{20}$

D) $\frac{1}{4}$

146

If $6x - 4 = 11$, what is the value of $18x - 12$?

A) 33

B) 11

C) 45

D) $\frac{5}{2}$

147

A number p is 160% greater than a positive number q. A number r is 35% less than p. A number s is 20% greater than r. The number s is how many times the number q?

A) 1.248

B) 1.690

C) 2.028

D) 2.450

148

The function f is defined by

$$f(x) = \frac{9 - 4x}{7}.$$

What is the slope of the graph of $y = f(x)$ in the xy-plane?

A) $\frac{9}{7}$

B) $\frac{4}{7}$

C) -4

D) $-\frac{4}{7}$

149

Line k is defined by $4x - 6y = 17$. Line j is perpendicular to line k in the xy-plane. What is the slope of line j?

A) $\frac{2}{3}$

B) $\frac{3}{2}$

C) $-\frac{3}{2}$

D) $-\frac{2}{3}$

150

What is the y-coordinate of the y-intercept of the graph of $y = 5(3)^{x+2}$ in the xy-plane?

A) 5

B) 15

C) 45

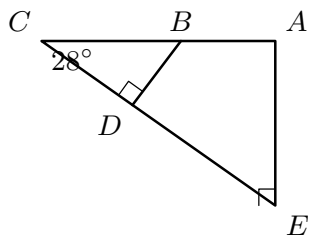
D) 135

151

For the positive values j and k, the ratio of j to k is 9 to 5. If j is increased by 36, by how much must k be increased so that the ratio of the new values of j to k is still 9 to 5?

Javob: _____

152



Note: Figure not drawn to scale.

In the figure shown, triangle CAE is similar to triangle CBD, where B lies on line segment CA and D lies on line segment CE. The measure of angle BCD is $28°$, angle CDB is a right angle, and $AE = 7(BD)$. What is the measure of angle CAE?

A) $28°$

B) $62°$

C) $196°$

D) $90°$

153

The expression

$$(6x - 2)(3x + 7) - (x - 4)^2$$

can be written in the form $ax^2 + bx + c$, where a, b, and c are constants. What is the value of $a + b + c$?

Javob: _____

154

An angle has a measure of $\frac{7\pi}{12}$ radians. What is the measure, in degrees, of an angle that is supplementary to this angle?

Javob: _____

155

A coach recorded the number of laps completed by each runner in two training groups during a practice session. The lists give the number of laps completed by each runner in each group.

Group R: 8, 14, 19, 23, 26, 31

Group S: 14, 19, 23, 26, 31

Which statement correctly compares the median number of laps for group R and the median number of laps for group S?

A) There is not enough information to compare the median number of laps for group R and the median number of laps for group S.

B) The median number of laps for group R is equal to the median number of laps for group S.

C) The median number of laps for group R is greater than the median number of laps for group S.

D) The median number of laps for group R is less than the median number of laps for group S.

156

$4x^2 + 14 = 32x + r$

In the given equation, r is a constant. The equation has exactly one real solution. What is the value of r?

Javob: _____

157

A laboratory culture of bacteria triples in size every 5 hours. The function

$$P(t) = 150(3)^{t/5}$$

gives the approximate number of bacteria in the culture t hours after the culture was started. Which statement is the best interpretation of $P(3 \cdot 5) = 4{,}050$ in this context?

A) The culture contains approximately 4,050 bacteria 3 hours after it was started.

B) The culture contains approximately 3 times 4,050 bacteria 15 hours after it was started.

C) The culture contains approximately 4,050 bacteria 3 five-hour periods after it was started.

D) The number of bacteria in the culture increases by approximately 4,050 every 5 hours.

158

The graph of the polynomial function f in the xy-plane, where $y = f(x)$, passes through the point $(6, -5)$. The function g is defined by $g(x) = f(x-2)+4$. Which of the following points must lie on the graph of $y = g(x)$?

A) $(8, -1)$

B) $(4, -1)$

C) $(8, -9)$

D) $(4, -9)$

159

The product of a positive number x and the number that is 9 less than twice x is 315. What is the value of x?

Javob: _____

160

Container P holds k liters of water. Container Q holds 72% of the amount of water in container P, and container R holds 150% of the amount of water in container Q. The positive difference between the amount of water in container R and the amount of water in container P is 32.4 liters. What is the value of k?

Javob: _____

161

For the linear function f, $f(-2) = 17$ and $f(4) = -1$. Which equation defines f?

A) $f(x) = -\frac{1}{3}x + \frac{49}{3}$

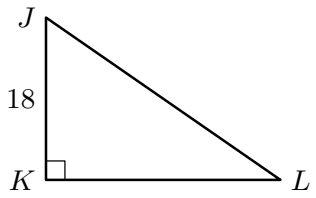
B) $f(x) = -3x + 23$

C) $f(x) = -3x + 11$

D) $f(x) = 3x + 23$

162

In right triangle JKL shown, angle K is a right angle and $JK = 18$. Which expression represents the length of line segment KL?



Note: Figure not drawn to scale.

A) $18 \tan L$

B) $\frac{18}{\cos L}$

C) $18 \sin L$

D) $\frac{18}{\tan L}$

163

A boat rental company charges \$95 for the first 2 hours of renting a boat and \$38 for each additional half hour, plus a fuel charge. Which equation represents this situation, where y is the total cost, in dollars, of renting a boat for x hours, f is the fuel charge, in dollars, and $x \geq 2$?

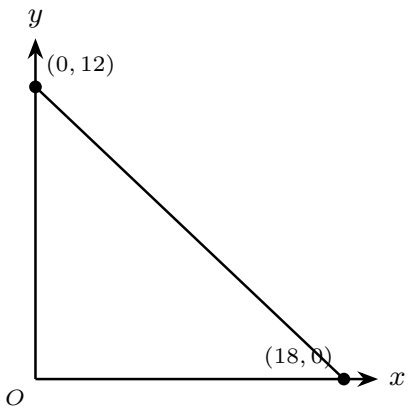
A) $y = 95 + 38(x - 2) + f$

B) $y = 95 + 76x + f$

C) $y = 95 + 76(x - 2) + f$

D) $y = 95 + 38(2x - 2) + f$

164



The line models the possible combinations of the number of small cakes, x, and the number of large cakes, y, that a baker can decorate in 720 minutes. It takes the baker a fixed amount of time to decorate each small cake and a different fixed amount of time to decorate each large cake. How many more minutes does it take the baker to decorate each large cake than to decorate each small cake?

A) 6

B) 20